Climate and Digital Indicator Replication Package

This README summarizes the replication package for two annual-report text indicators: climate-risk attention and corporate digital transformation. The package uses an AEA/AER-style structure with a master script, modular code, dictionaries, sample data, output folders, logs, and source-code provenance.

1. Data Availability Statement

The full empirical data are not included. The code is designed to run on annual-report PDFs for Chinese listed firms. Users should obtain annual reports from legitimate public disclosure sources and comply with those sources' terms. The sample PDF in data/sample/ is synthetic and only for testing.

2. Main Contents

```
run_all.py                          Master script
code/01_convert_pdf_to_txt.py         PDF-to-TXT conversion
code/02_compute_climate_attention.py  Climate-risk attention
code/03_compute_digital_transformation.py
                                    Digital-transformation indicator
code/04_merge_indicators.py           Merge firm-year indicators
code/05_build_dyadic_variables.py     Build dyadic divergence variables
data/dictionaries/                    Keyword dictionaries
data/sample/                          Synthetic sample PDF and metadata
legacy/                               Original source-code materials
```

3. Software

Recommended: Python 3.10+, pdfminer.six, PyMuPDF, pypdf, jieba. The jieba package is recommended for closest replication of the referenced Chinese text-analysis code. reportlab is needed only to regenerate the synthetic sample PDF.

```
pip install -r requirements.txt
```

4. Sample Test

```
python run_all.py \
  --pdf-dir data/sample \
  --metadata data/sample/report_metadata.csv \
  --txt-dir data/intermediate/sample_txt \
  --out-dir data/output/sample \
  --overwrite
```

The main sample output is data/output/sample/firm_year_text_indicators.csv and should contain one firm-year observation for 000001, SampleFirm, 2022.

5. Full Workflow

```
# Step 1: put annual-report PDFs in:
#   data/raw/annual_reports_pdf/

# Step 2: convert PDF to TXT
python code/01_convert_pdf_to_txt.py \
  --pdf-dir data/raw/annual_reports_pdf \
  --txt-dir data/intermediate/txt \
  --log logs/01_pdf_to_txt_log.csv \
  --overwrite

# Step 3: climate-risk attention
python code/02_compute_climate_attention.py \
  --txt-dir data/intermediate/txt \
  --method auto \
  --output data/output/climate_attention.csv

# Step 4: digital transformation
python code/03_compute_digital_transformation.py \
  --txt-dir data/intermediate/txt \
```

```
  --output data/output/digital_transformation.csv

# Step 5: merge indicators
python code/04_merge_indicators.py \
  --climate data/output/climate_attention.csv \
  --digital data/output/digital_transformation.csv \
  --output data/output/firm_year_text_indicators.csv
```

6. Indicator Definitions

climate_attention_x100 = climate_keyword_count / total_word_count * 100. digital_transformation_log1p = log(1 + total digital keyword count). For customer-supplier analysis, climate_attention_divergence_x100 = customer_climate_attention_x100 - supplier_climate_attention_x100.

7. Provenance

legacy/climate_attention_original/ contains the original climate-risk text code copied from the supplied Wang-style replication materials. legacy/digital_transformation_original/ contains the Annualreport_tools scripts used for annual-report crawling, PDF-to-TXT conversion, and keyword analysis. The code/ directory contains the cleaned command-line version prepared for public replication.